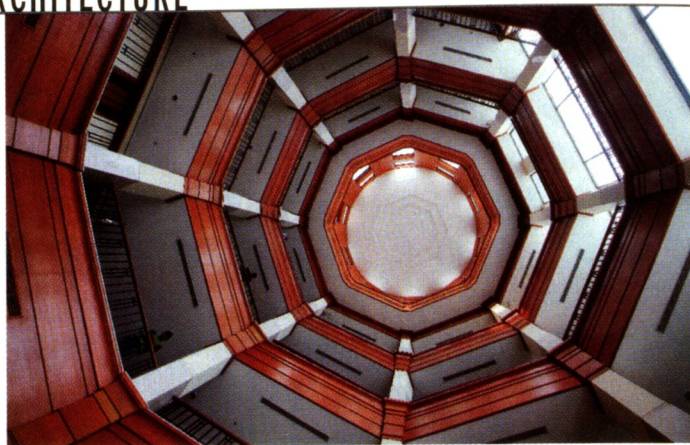


for Integrated Habitat Assessment (GRIHA) too. However, while many are developing green projects, not all are seeking LEED certification which involves a stringent list of do's and don'ts. "The idea is to quantitatively measure how much energy a project can save by following certain guidelines," says Milli Majumdar, associate director, Sustainable Building Science, The Energy and Resources Institute (TERI). However, with lack of uniform standards, the industry is divided between

RAY OF HOPE: ITC's Green Centre in Gurgaon (below) uses natural lighting in the day (right) which has reduced its energy consumption by around 51 per cent



ECO-FRIENDLY PRODUCTS

LESS VOLATILE ORGANIC COMPOUNDS: Less toxic and less hazardous	Asian Paints
WATERLESS URINAL: Saves water	E.I.D. Parry
DOUBLE GLAZED GLASS: Saves energy costs by as much as 30%	Asahi India and St. Gobain
CFC-FREE HVAC EQUIPMENT: Uses refrigerants which have low ozone depletion potential	Blue Star and Carrier

(CFC: chloro fluoro carbon; HVAC: heating, ventilation and air conditioning)

the LEED and TERI's GRIHA ratings, both of which use different approaches to achieve the same goal.

But simply changing your light-bulbs is not enough. Green architecture involves careful planning. "It brings huge amounts of discipline in your construction process," says Niranjana Khatri, head of ITC's clean development division. A green building may not differ from any other in appearance, but the difference lies in the fundamentals. But, being a new concept, such projects involve a system that designers and labourers are working with for the first time, says Tanmay Tathagat, principal, Environmental Design Solutions, a green building consultancy. "Every project is a new experience," he says.

When the ITC Green Centre was being constructed in Gurgaon in 2006, the biggest challenge was following what was relatively a new concept. Hundreds of issues—ranging from heat

and water management to controlling "light pollution" to reusing old furniture had to be taken into account. However, it wasn't easy. In its bid to be 'platinum', the company had to follow LEED standards which had been developed as per the American framework. For instance, it required a petrol pump or an alternate source of fuel on the company's premises. This was not possible under Indian regulations. "We asked CII to Indianise the norms, which it did," says Khatri, suggesting "more points should be given to critical issues, such as water management." However, the company did follow many requirements including capped lighting (which means that it should not be reflected in the atmosphere to avoid disturbing the birds) or using 10 per cent wood from a certified source (which replants trees for those chopped). Today ITC has reduced its energy and water consumption by 51 per cent and 40 per cent, respectively. With lower utility bills, Khatri expects to break even (on an extra cost of 12 per

cent) within five years and then realise subsequent savings.

All this has spurred suppliers to produce green equipment and materials that can save energy. For instance, "a big contributor to heat gain in a building is the kind of glazing used," says Kunwar Narayan, COO, AIS Glassolutions. "Using double glazed glass can reduce heat gain by four times." The potential for these products, estimates IGBC, could be around Rs 1,600 crore by 2010. The supply of raw materials and services is going up and the price differential (between a regular and green building) is coming down which indicates that the trend is here to stay. Just a few years ago, green construction could cost 10-15 per cent or more than standard construction. Today that margin is far lower. Overall, green buildings offer a plethora of benefits. They are healthier, improve productivity, and make for happier employees. They also save energy. Who wouldn't want that? ■